

StepUp App

Project Proposal for UCS503P

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August 18, 2026

1 Higher-order goal

...secondary framing

This project aims to support healthier, more socially connected student life on campus by making physical activity tracking trustworthy and by making it easier for students to find reliable sport partners. While the topic is broader than any single feature, the immediate engineering objective is to translate trustworthy activity tracking and effective sport-group formation into a measurable, deployable software solution.

2 Time-to-value

...primary heuristic

- We will deliver meaningful increments quickly by prototyping the core user workflow of StepUp and validating it with users within a short iteration window.
- We will prioritize fast feedback over perfection by using weekly iterations (and targeting CI-backed automation early) to reduce release friction.
- Success is defined by what can be delivered and measured in the first iteration, then improved based on user feedback.

3 Problem Statement

Campus fitness and sport-partner discovery on Thapar's campus currently suffer from two distinct gaps:

- **Unverified activity data:** existing fitness-tracking approaches trust imported step and activity data without checking whether it reflects genuine walking or running. Records may instead come from phone shaking, duplicate syncs, or movement while travelling in a vehicle, which undermines any challenge or leaderboard built on top of that data.
- **Sport-partner discovery as a filtered list, not a match:** students looking for regular sport partners or groups (e.g., badminton, football, running) currently rely on informal channels such as WhatsApp groups, rather than a system that actually matches them on sport, skill, availability, group size, location, and reliability.

Why this matters now (contextual relevance): Any gamified fitness or matchmaking feature is only as useful as the trust students place in it; unverified activity data and ad-hoc partner discovery both directly undermine that trust, which motivates addressing both problems as first-class engineering concerns rather than afterthoughts.

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4 Proposed Solution

...Software Proposition

4.1 Overview

Build the StepUp App, consisting of an Android application and a web dashboard, with the following components:

- **Android application (student-facing):** activity synchronization via Health Connect, an on-device activity-integrity check, buddy matching requests, and challenge participation.
- **Activity-integrity engine:** distinguishes genuine walking/running from phone shaking, duplicate/repeated sync records, and vehicle movement, running mostly on-device.
- **Sport group matching:** treats sport-buddy formation as a group-formation/matching problem based on sport preference, skill level, availability, group size, coarse location, and reliability, rather than a filtered contact list.
- **Web dashboard:** challenge configuration, pilot monitoring, and aggregate analytics (participation trends, integrity-flag rates, match success rates).

4.2 Core workflow

1. Student links activity data through Health Connect on the Android app.
2. The on-device activity-integrity check evaluates incoming activity records (genuine walking/running vs. phone shaking, duplicate sync, or vehicle movement) before they count toward a challenge.
3. Student optionally submits a sport-buddy request specifying sport, skill level, available time slots, preferred group size, and coarse location.
4. The backend matching engine proposes a group or partner based on these constraints and each student's reliability history.
5. Admins/organizers use the web dashboard to configure challenges, monitor the pilot, and review aggregate analytics.

4.3 Operational constraints for a campus setting

- Activity data will be synchronized via Health Connect (not raw continuous GPS tracking).
- The activity-integrity check should run primarily on-device to minimize data transmission and enable near-real-time feedback.
- Matching will use only coarse location (e.g., locality/hostel block), not precise GPS coordinates.
- The Android application will be built using Kotlin.

5 Solution Approach

...Engineering focus

This is an engineering course project, so we focus on scalable implementation, workflow correctness, and rigorous evaluation against simple baselines, rather than claiming state-of-the-art research contributions.

5.1 Activity-integrity assistance

...non-research

Activity-integrity detection will be developed and evaluated against a simple baseline:

- Build a labelled dataset using short recorded sessions covering genuine walking, genuine running, phone shaking/fake steps, duplicate or repeated sync records, and movement while travelling in a vehicle.
- Use a rule-based baseline with thresholds on step cadence, variance, and speed without claiming state-of-the-art.
- Develop the proposed detector and compare it against the rule-based baseline on held-out recorded sessions.

5.2 System architecture

...solution architecture

A typical multi-component architecture:

- **Android application:** built in Kotlin; handles Health Connect activity synchronization, the on-device activity-integrity check, buddy matching requests, and challenge participation.

- **Web dashboard:** supports challenge configuration, pilot monitoring, and aggregate analytics, including participation trends, integrity-flag rates, and match success rates.
- **Backend:** hosts the matching engine, exposes APIs consumed by both the Android application and the web dashboard, and performs server-side integrity-model comparison and evaluation logging.

5.3 CI/CD and fast delivery enabler

CI/CD will be used to:

- Automatically build and run tests on every change.
- Produce repeatable deployment artifacts to staging.
- Enable safe and frequent releases of incremental features.

6 Evaluation Criterion

...measurable and attributable

We define success using measurable metrics tied to the two core problems: activity integrity and sport group matching.

6.1 Primary evaluation metric

F1 score of the proposed activity-integrity detector, compared against the rule-based baseline, measured on our labelled dataset.

- **Target:** the proposed detector should achieve a higher F1 score than the rule-based baseline. Specific target values will be set once baseline results are available.
- **Attribution:** computed by comparing detector predictions against ground-truth labels on held-out recorded sessions.

6.2 Secondary evaluation metrics

- **Activity integrity:** precision, recall, false-positive rate, on-device latency, and battery cost of running the detector.
- **Matching quality:** match acceptance rate, fairness across users, match stability, and successful completed sessions versus formed matches, each compared against random matching and first-fit matching baselines.
- **Participation:** repeat participation observed in weeks 2–3 of the pilot compared with week 1.

6.3 Pilot validation plan

...high-level

- Recruit volunteers from our batch/department and hostel/class groups, targeting approximately 40–60 students.
- Run the pilot for 2–3 weeks, covering at least one complete challenge cycle.
- Track activity-integrity flags, match acceptance/no-shows, and repeat participation throughout the pilot window to evaluate the metrics above.

7 Scalability

...with foresight

The proposition should scale in both theoretical and practical senses.

1. **Scalable (theoretically):** The system should support growth in number of students, challenges, and concurrent matching requests by:
 - decoupling the Android application, web dashboard, and backend,
 - enabling pagination and indexing for common queries (e.g., challenge participation, match history),
 - using stateless API design for the backend.
2. **Operationally deployable (practically):** The system should run on commonly available hosting/deployment infrastructure:
 - managed database,

- containerized or platform-hosted backend deployment,
- CI/CD pipeline for staging/production.

8 Engine availability heuristic

A mature set of tools/services is available to implement this as a campus software project:

- Health Connect API for activity data synchronization on Android.
- Kotlin/Android SDK for the student-facing application.
- A backend web framework for REST APIs (specific choice to be finalized).
- A managed relational or document database (specific choice to be finalized).
- An authentication approach (token-based or session-based, to be finalized).
- A test framework for unit/integration tests.
- A CI runner and staging deployment target.

We will use these mature components to focus on the activity-integrity and matching problems, and on measurement, rather than inventing new infrastructure.

9 Project scope and deliverables

...iteration-friendly

9.1 Initial deliverable

...first iteration

- Labelled activity-integrity dataset (genuine walking, genuine running, phone shaking, duplicate sync, vehicle movement) and rule-based baseline.
- Android application skeleton with Health Connect synchronization and challenge participation flow.
- Initial matching formulation with random and first-fit baselines.
- Basic logging for activity-integrity and matching metrics.
- CI: build + backend unit tests + linting.
- CD to staging with a single command or pipeline job.

9.2 Subsequent deliverables

- Proposed activity-integrity detector, integrated on-device, and its evaluation against the rule-based baseline.
- Refined sport-group matching approach, evaluated against random and first-fit baselines.
- Web dashboard for challenge configuration, pilot monitoring, and aggregate analytics.
- Privacy controls: consent flow, data/match-history deletion, and visibility controls.
- Execution of the campus pilot and reporting of results against the evaluation metrics.

10 Risks and mitigations

- **Data privacy / consent:** require explicit consent before collecting activity or location data; do not store continuous GPS trails and use only coarse location for matching; process and discard raw sensor data on-device where possible; provide data deletion, match-history deletion, and visibility controls so students choose what matched buddies can see.
- **User adoption:** keep the on-device integrity check unobtrusive and the matching request flow simple, so participation in challenges and buddy matching requires minimal friction.
- **Evaluation measurement ambiguity:** finalize the labelled-dataset protocol and define activity-integrity and matching metrics before development begins, to avoid ambiguity when comparing against baselines.

- **Operational issues in pilot:** use staging early; ensure CI/CD produces repeatable deployments; test Health Connect synchronization across a range of devices before the pilot begins.

11 Summary

This project proposes the StepUp App, a campus fitness and sports-partner platform for Thapar students that addresses two concrete problems: verifying whether imported activity data is genuine, and treating sport-buddy formation as a real matching problem rather than a filtered list. It meets the course criteria by emphasizing time-to-value through rapid iterations, implementing CI/CD to support frequent safe releases, and using measurable evaluation criteria for both the activity-integrity detector (compared against a rule-based baseline) and the matching approach (compared against random and first-fit baselines), validated through a campus pilot of approximately 40–60 students.